

Manavpreet Singh

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EDUCATION

University of Calgary

Sept 2023 - April 2028

Combined Degree: B.Sc Computer Science and B.Comm. Business Technology Management

Calgary, Alberta

Relevant Coursework: Information Technology Ethics, Software Engineering, Computing Machinery, Information Security and Privacy, Data Structures and Algorithms, Statistics.

EXPERIENCE

Machine Learning and Applied AI Engineering Co-op Student

May 2026 – Present

Health Shared Services : Web and Collaboration Team, Alberta Health Services

Calgary, Alberta

- Designed and prototyped a cross-encoder reranker for AVA, AHS's internal **AI-powered RAG assistant** supporting Alberta Health Link 811 triage nurses across 1,000+ healthcare resources, improving top-ranked document retrieval in the production search workflow.
- Built reranker evaluation pipelines using **Python, PyTorch and Hugging Face Transformers**, improving top-3 retrieval performance by **56%** relative lift in proof-of-concept testing compared to baseline vector search.
- Benchmarked **MiniLM** and **BGE-reranker-base** across ranking accuracy, latency, and model size, identifying lightweight reranker candidates to improve relevance while collaborating in daily standups with the **AI Integration team**.
- Integrated AVA's existing **vector search API** into the reranker evaluation workflow, retrieving top-25 candidate documents and measuring whether cross-encoder reranking could move expected clinical resources into the top-3 results.

Machine Learning and AI Research Intern

Dec 2025 – Present

Data and Network Sciences and Applications Lab, University of Calgary

Calgary, Alberta

- Applied TabPFN based **machine learning models** to forecast **chronic disease** trends from longitudinal health datasets and **Python-based analytics** (Numpy, scikit-learn) to inform resource allocation for **Alberta Health Services**.
- Assisted in **data analysis** and reporting by writing **SQL queries** to **extract and validate structured datasets** and built **Power BI dashboards** to visualize trends and support operational decision-making.
- Conducted **data quality assessments** and reconciliations to maintain and improve **database accuracy** by detecting and addressing inaccuracies, missing entries, and duplicate records, ensuring **reliable reporting and documentation**.
- Created and maintained detailed **documentation and analysis** of **data workflows**, standards, and validation outcomes.

Information Technology Lab Supervisor

Jan 2026– Apr 2026

AFC Lab, University of Calgary

Calgary, Alberta

- Delivered **Tier 1 technical support** for **Windows 11** and **macOS lab environments**, diagnosing **hardware, software, and access-related issues** to ensure uninterrupted academic operations.
- Assisted in management of 100+ Windows-based workstations, ensuring compliance with institutional IT policies and baseline security standards..
- Managed **concurrent technical incidents** across **multiple lab environments**, **triaging issues** by severity to maintain **service-level response times** and **operational continuity**.

PROJECTS

[Online Multiplayer Game Platform](#) | *Python, FastAPI, React.js, MongoDB, pytest, Jest, Git*

Jan 2026 – Apr 2026

- Built a **full-stack online multiplayer gaming platform** using **Python (FastAPI), React.js, and MongoDB**, supporting user authentication, profile management, matchmaking, lobby-based game rooms, and live turn-based gameplay, as part of a **25-person Agile team** simulating an industry-style product workflow.
- Led automated testing writing **pytest backend test suites** and **Jest frontend tests** to validate platform flows including matchmaking, lobby management, turn progression, and game-end handling.
- Built **integration tests covering end-to-end user flows** (login → lobby → matchmaking → turn → game-end) to catch regressions before merges and keep the main branch in a runnable state throughout the 4-month development cycle.
- Extended contributions beyond QA into: **FastAPI endpoints** for room/lobby lifecycle and turn progression on the backend, **React components for lobby and in-game UI** on the frontend, and **UML class + use case diagrams** to align cross-sub-team architecture decisions.
- Practiced industry-style Git workflows**: feature branches, pull requests, peer code reviews, and incremental merges — within a team operating under Scrum.

[Sleep Tracker App](#) | *Java, JUnit, CSV, GitHub*

- Built a **Java-based sleep tracking application** that logs nightly duration, quality ratings, and lifestyle factors (caffeine, exercise, screen time), persisting **30+ data points per entry** to CSV for offline-first analysis.
- Architected the backend around the **Model-View-Controller pattern** with abstract base classes for sleep entries and lifestyle factors, enabling new tracking metrics to be added without modifying existing analysis logic; implemented **7-day rolling averages** and correlation analysis between lifestyle factors and sleep quality.
- Achieved **85%+ unit test coverage** with JUnit across edge cases including malformed CSV rows, leap-year date handling, and concurrent file access, catching **12+ bugs** before integration.

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, JavaScript, Rust, TypeScript, Go

AI/ML & Data Science: Power BI, PyTorch, Hugging Face, RAG, Vector Search, NumPy, pandas, scikit-learn, matplotlib

Cloud & DevOps : AWS (S3, Transcribe, EC2, Lambda), Docker, Kubernetes, GitHub Actions

Frameworks & Libraries: React.js, Node.js, FastAPI, Django, Flask, Tailwind CSS

Databases: PostgreSQL, MongoDB, Firebase, Supabase, Redis

Tools & Testing: Microsoft Office 365, Git, VS Code, Jupyter Notebook, Google Colab, Jest, Selenium, JUnit, Android Studio, RStudio